Received: April 1, 2026. Revised: May 5, 2026. 28

International Journal of
Intelligent Engineering & Systems

http://www.inass.org/

Residual TCN-BiLSTM with Attention Framework for Epileptic Seizure Detection from EEG Signals

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Abstract: Epileptic seizures occur due to aberrant brain electrical activity and significantly impact the patient's overall health and well-being. Early and reliable identification of epileptic seizures plays a crucial role in improving diagnosis and guiding appropriate treatment. Electroencephalogram (EEG) signals are widely used to monitor brain activity and provide important information for identifying seizure patterns. However, the detailed temporal dynamics and noise present in EEG signals make automated seizure detection a challenging task, often leading to high false positive rates and missed detections in clinical settings. This study proposes a Residual Temporal Convolutional Network-Bidirectional Long Short-Term Memory (LSTM) with Attention (Residual TCN-BiLSTM-Attention) architecture that embodies both hierarchical temporal patterns and long-range dependencies in EEG signals. The proposed framework combines residual temporal convolution to extract temporal features from EEG sequences, BiLSTM to capture long-range relationships in forward and backward directions, and an attention technique to highlight the most useful EEG characteristics for categorization. EEG recordings from the Mendeley EEG dataset, consisting of multichannel seizure and non-seizure brain signals, were utilized to evaluate the proposed approach. The EEG data were preprocessed through normalization and segmentation before being used for model training and evaluation. Experimental results show that the proposed framework achieves a 98.8% accuracy, along with high recall, precision, F1-score, and AUC values, indicating effective classification performance. Comparative analysis with existing methods further demonstrates competitive performance of the proposed model under the same evaluation conditions. These findings highlight the capability of the proposed hybrid architecture in capturing discriminative temporal EEG patterns for seizure detection. However, the results are obtained under a segment-level evaluation protocol and reflect performance specific to the defined experimental setting.

Keywords: Electroencephalogram, Deep learning, Bidirectional long short-term memory, Epileptic seizure detection, Temporal convolutional network.

1. Introduction

The long-term neurological condition known as epilepsy is characterized by regular, spontaneous seizures that occur due to abnormal electrical impulses in the brain. It affects millions of individuals globally and is considered a highly prevalent disorder of the nervous system. EEG is widely utilized for monitoring brain activity and plays a crucial role in epilepsy diagnosis because it records electrical signals generated by neurons through electrodes placed on the scalp. EEG provides detailed temporal information about brain activity and enables clinicians to observe seizure patterns and abnormal neural behavior. However, the interpretation of EEG signals remains difficult because of their complexity, high variability, and non-stationary nature. EEG recordings are frequently affected by artifacts generated from eye movements, muscle contractions, and external environmental noise, which makes accurate seizure detection more challenging. Moreover, the transition between normal brain activity and seizure activity (ictal state) is often subtle, making manual analysis difficult and time-consuming [1, 2].

Worldwide, epilepsy affects around 50 million individuals, and a substantial proportion of these patients live in developing countries where

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specialized neurological care is often insufficient or difficult to access. Delayed diagnosis or misinterpretation of EEG recordings can lead to ineffective treatment approaches and reduce the patients' quality of life. In addition to physical wellness risks, epilepsy is also associated with psychological distress, injury risks, and social stigma. Therefore, continuous monitoring and timely identification of seizure events are essential for effective epilepsy management. The growing prevalence of neurological disorders and the demand for reliable diagnostic tools have motivated the development of automated EEG-based seizure detection systems [3, 4].

Over the years, several computational approaches have emerged to automate seizure detection from EEG signals. Traditional methods typically rely on signal processing techniques to extract discriminative features from EEG recordings. Methods such as the Discrete Wavelet Transform and Fourier Transform are frequently applied to analyze EEG signals in both time and frequency domains. Although these approaches contribute to automated seizure detection, they depend heavily on handcrafted features and predefined signal processing procedures. Consequently, such methods struggle to record the composite nonlinear patterns inherent in EEG signals, which limits their effectiveness in real-world clinical environments [5].

Recent advances in Artificial Intelligence (AI), DL and machine learning (ML) have significantly improved automated EEG analysis. DL techniques are able to automatically learn hierarchical feature representations from complex biomedical signals and can capture both spatial and temporal dependencies within EEG recordings, thereby improving seizure classification performance [6, 7]. However, some existing DL architecture struggle to efficiently model long-range temporal dependencies present in EEG sequences and often need complex architectures or extensive preprocessing steps to attain reliable detection performance. In addition, many approaches emphasis primarily on feature extraction while providing incomplete capability for highlighting the most informative temporal segments associated with seizure events. These limitations have motivated the development of a Residual TCN-BiLSTM-Attention hybrid architecture that integrates complementary learning mechanisms for improved temporal modeling and discriminative feature representation in EEG signals. The primary goals of the present study are summarized as follows:

- A hybrid Residual TCN–BiLSTM–Attention architecture is developed for modeling temporal EEG patterns associated with epileptic seizures.
- The proposed framework is evaluated on a publicly available EEG dataset, demonstrating strong classification performance under the defined experimental setup.
- A comparative analysis is conducted to examine the effectiveness of the proposed architecture relative to existing approaches within the same evaluation setting.

The structure of the remaining sections is as follows: The existing literature on automated epileptic seizure identification is reviewed in Section 2. In Section 3, the study outlines the materials and methods employed and the proposed hybrid model architecture. A detailed analysis of the experimental results, along with comparisons with existing methods, is provided in Section 4. Lastly, summary of the major findings and recommendations for future areas of investigation is presented in Section 5.

2. Related works

Paneru [8] developed a multi-dimensional CNN-Bi-GRU for epileptic seizure recognition by means of EEG from the Mendeley dataset. The study integrated CNN and Bi-GRU for temporal pattern learning in multichannel EEG signals. Performance evaluation showed an accuracy of 97.43%, outperforming benchmark architectures such as Electroencephalography (EEGNet) Network with 92.17% accuracy and Convolutional Temporal Network (CTNET) with 85.11% in seizure versus non-seizure classification. The main limitation was offline evaluation with limited seizure-class training samples and computational latency during deployment on Raspberry Pi hardware.

Alzamili et al. [9] developed a clustering-based EEG analysis framework using Ruzicka mathematical resemblance combined through Particle Swarm Optimization (PSO) for feature selection to improve epilepsy classification. The method grouped EEG records into two clusters according to the Ruzicka similarity measure, enabling the formation of coherent data groups for efficient analysis. After clustering, PSO identified optimal feature subsets to reduce dimensional complexity while preserving discriminative information. Experimental evaluation was conducted by means of the Bonn University EEG dataset and classification performance was assessed using performance metrics. The main limitation was reliance on traditional classifiers, reducing generalization across diverse EEG recordings.

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Darankoumal et al. [10] designed a DL-based model for epileptic seizure recognition by means of EEG signals. The study investigated automated seizure detection pipelines by integrating CNN with a transformer encoder to capture spatial and temporal characteristics of EEG recordings. Preprocessing and post-processing stages were incorporated to support realistic clinical scenarios and improve detection reliability. The model was trained by means of EEG data recorded from epileptic mice and evaluated on human EEG recordings to assess cross-domain generalization capability. Experimental evaluation demonstrated an F1-score on human EEG data, indicating effective detection performance across heterogeneous datasets. The main limitation was reliance on animal-based training data and differences between preclinical and clinical EEG recordings, which restricted broader clinical applicability.

Ikizler and Ekim [11] conducted a study on automatic epileptic seizure detection using EEG signals by integrating spectral, statistical, and wavelet-based features. The selected features were assessed by means of multiple ML models with RF, KNN, SVM, Gradient Boosting, and Deep Neural Network (DNN) within a hybrid evaluation framework. Experiments were performed on multiple class combinations to assess classification capability across different seizure scenarios. Results demonstrated high classification performance, with the DNN achieving superior detection capability among the evaluated models. The main limitation was dependence on handcrafted feature extraction and complex feature engineering, which restricted the method's capability to automatically learn deep temporal EEG patterns.

Atitallah et al. [12] developed a DL framework for epileptic seizure prediction by means of EEG signals by integrating a Graph Convolutional Network (GCN) with a Kolmogorov-Arnold Network (KAN) layer within a Self-Supervised Learning (SSL) framework. EEG signals were transformed into graph representations to capture spatial relationships among channels and enable effective extraction of complex nonlinear patterns. The SSL strategy facilitated representation learning from unlabeled EEG data, reducing dependence on large annotated datasets. Experimental evaluation was performed using the CHB-MIT EEG dataset to assess seizure prediction capability across patients. Results demonstrated improved seizure prediction performance with high classification metrics, indicating effective generalization across EEG recordings. The main limitation was dependence on graph transformation and dataset-specific training, which restricted broader applicability across heterogeneous EEG datasets.

Shawly and Alsheikhy [13] developed an attention-enhanced DL method for epileptic seizure detection by means of EEG signals. The method integrated a Convolutional Neural Network (CNN) with a Novel Attention Module (NAM) to enhance feature extraction from EEG recordings. Fourier Transform (FT) was utilized for frequency-domain feature extraction, while Principal Component Analysis (PCA) reduced dimensional complexity before classification. Model training and evaluation were performed using 3 publicly available EEG datasets obtained from PhysioNet and ResearchGate sources. Experimental evaluation applied metrics together with sensitivity, F1-score and specificity to assess seizure detection efficacy across different configurations. Results demonstrated high classification capability with improved interpretability through SHapley Additive exPlanations (SHAP). The main limitation was dependence on multiple heterogeneous datasets and extensive preprocessing steps, which increased computational complexity and reduced adaptability across unseen EEG recordings.

Betgeri et al. [14] designed a hybrid seizure prediction framework using EEG signals by integrating XGBoost with RNN constructions including LSTM and Gated Recurrent Unit (GRU). Signal preprocessing techniques were applied to improve EEG data quality before model training. Sixteen ML and DL methods for classification using the same EEG dataset of epileptic patients. Experimental evaluation demonstrated improved predictive performance, where the hybrid XGBoost-LSTM-GRU architecture achieved an 96.8% accuracy with high recall for seizure recognition. The main limitation was dependence on larger and various EEG datasets for improving generalization across different patient demographics and seizure patterns.

Khan et al. [15] developed an EEG-based seizure detection framework by integrating ML classifiers with explainable AI techniques. The method included preprocessing using band-pass filtering and epoch segmentation, followed by statistical feature extraction to differentiate seizure and non-seizure EEG patterns. Classification was performed using SVM, KNN, and RF algorithms. Experimental evaluation was conducted by means of the CHB-MIT scalp EEG dataset, which comprises recordings from 23 subjects including epileptic patients and healthy controls. Performance analysis indicated that the RF classifier achieved the higher classification efficiency through improved recall, precision, and F1-score

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compared with other models. Interpretability of predictions was enhanced by means of SHAP and Local Interpretable Model-Agnostic Explanations (LIME). The main limitation was reliance on handcrafted statistical features and traditional machine learning models, limiting automated learning of complex EEG patterns.

Tchinda et al. [16] introduced a DL-based epileptic seizure detection framework by means of EEG signals transformed into log-Mel spectrogram representations. The method converted EEG signals into image-based features by applying logarithmic Mel-scale spectrogram extraction, enabling CNN to perform classification on the generated spectrogram images. Experimental evaluation was performed by means of the Bonn EEG dataset and the NSC-ND dataset. Performance analysis demonstrated high classification capability, along with improved sensitivity, specificity, and F1-score across evaluations. The main limitation was dependence on spectrogram-based feature transformation and limited dataset diversity, which reduced adaptability across heterogeneous EEG recordings.

Malakouti et al. [17] developed an ML-based epileptic seizure detection framework using EEG signals combined with bandpass filtering and Discrete Wavelet Transform (DWT) used for time-frequency feature extraction. Hyperparameter tuning through grid search and ensemble techniques was incorporated to enhance classification performance. Experimental evaluation was conducted using the Bonn EEG dataset to assess seizure detection capability. Results indicated that the ARTMAP classifier with preprocessing and ensemble strategies attained the maximum efficiency with an AUC of 0.85 and accuracy of 85.3%, demonstrating effective classification with computational efficiency. The main limitation was dependence on handcrafted feature extraction and adaptive ML classifiers, which restricted automated learning of complex EEG patterns compared with DL-based architecture.

Bhadra et al. [18] designed a DL-based epileptic seizure identification framework using EEG signals by integrating a hybrid architecture composed of a one-dimensional CNN followed by two GRU layers for feature learning and classification. The framework, named HyEpiSeiD, extracted discriminative features from preprocessed EEG signals and performed seizure classification through the DL pipeline. Experimental evaluation was conducted using 2 widely existing datasets, namely the Epilepsy dataset UCI and the Mendeley EEG dataset. Performance analysis demonstrated higher classification accuracy on the UCI dataset and Mendeley dataset. The main limitation was reliance on preprocessed EEG datasets, which restricted evaluation on raw EEG signals and reduced generalization across heterogeneous clinical recordings.

Kina et al. [19] developed an ML- and DL-based epileptic seizure. The classification framework employed RNN, CNN, ERT (Extremely Randomized Trees), and ResNet (Residual Neural Network) algorithms to identify seizure patterns from EEG recordings. Experimental evaluation was achieved by separating the dataset as training and testing subsets and applying 10-fold cross-validation for performance assessment. Outcomes indicated that the RNN achieved the best efficacy with an overall accuracy of approximately 97% in epileptic seizure classification. The main limitation was the absence of evaluation on imbalanced datasets and limited validation across heterogeneous EEG recordings, which restricted generalization capability.

Shah et al. [20] developed an ML-based epileptic seizure detection framework using EEG signals by integrating feature and channel reduction techniques with Explainable AI. Temporal-domain features were extracted from EEG recordings using a one-second time window, and only six discriminative features from five EEG channels were selected for classification. The framework employed XGBoost as the ML classifier and utilized SHAP to analyze feature importance and improve model interpretability. Experimental evaluation was conducted using the University of Beirut Medical Center (UBMC) EEG dataset containing records from 6 patients with ictal and interictal segments acquired at a sampling rate of 500 Hz using the international 10-20 electrode system. Results demonstrated performance exceeding 95% across evaluation metrics for seizure detection. The main limitation was the small dataset containing only six patients, which restricted model generalization across broader clinical populations.

Li et al. [21] designed a DL-based epileptic seizure detection framework using EEG signals by integrating a CNN with a Vision Transformer (ViT) through a Multi-Stream Feature Fusion (MSFF) strategy. EEG signals were transformed into time-frequency representations by means of the TQWT to enable effective feature extraction from both local and global temporal dependencies. The CNN module captured spatial EEG characteristics, while the ViT module modeled global time-series relationships for seizure classification. Experimental evaluation was conducted using the CHB-MIT EEG dataset and the Kaggle 121-people epilepsy dataset. Results demonstrated a classification accuracy of 98.85% on the CHB-MIT dataset, indicating effective seizure

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recognition performance and improved generalization capability. The main limitation was dependence on image-based EEG transformation and evaluation on limited datasets, which restricted validation across diverse clinical EEG recordings.

Shi et al. [22] developed an ML-based epileptic seizure detection framework using EEG signals by introducing a sample similarity-based feature selection strategy to reduce redundant features while preserving classification effectiveness. Multi-view EEG features from temporal, spectral, and time-frequency domains were analyzed to construct feature representations for seizure recognition. The method improved intra-class similarity and decreased inter-class similarity to select discriminative features through a heuristic search strategy, thereby reducing feature dimensionality and improving model stability. Experimental evaluation was carried out with the help of Bonn EEG dataset and further validation was performed on the CHB-MIT dataset to assess model adaptability across different datasets. Results demonstrated improved classification accuracy, feature stability, and enhanced generalization capability associated with conventional feature selection methods. The main limitation was the restricted evaluation on limited datasets and seizure types, which reduced validation across broader clinical EEG populations.

2.1 Research gap

Although significant progress has been achieved in epileptic seizure detection using EEG signals, several methodological limitations remain in existing approaches. Many studies employ CNN and RNN combinations to obtain spatial and temporal EEG features; however, these architectures often struggle to effectively obtain long-range temporal dependencies within complex EEG sequences and rely heavily on hierarchical convolutional operations for feature extraction [8, 13, 18]. Several frameworks also depend on handcrafted feature extraction techniques or traditional machine learning classifiers, which limit the capacity of methods to learn deep nonlinear representations from raw EEG signals and reduce adaptability to varying signal characteristics [9, 11, 17, 19]. In addition, some approaches incorporate specialized transformations such as graph representations, spectrogram conversions, or feature clustering mechanisms that introduce additional computational complexity and increase model design dependency on preprocessing pipelines [12, 16, 21]. Other studies emphasize spatial or frequency-domain feature extraction but provide limited capability for modeling long-range temporal relationships and contextual dependencies inherent in EEG signals during seizure progression [10, 22]. Consequently, there remains a need for an efficient DL architecture capable of capturing long-range temporal dependencies while reducing reliance on complex preprocessing or handcrafted feature engineering.

3. Materials and methods

This study presents a hybrid DL architecture for epileptic seizure detection from EEG signals. The proposed framework integrates Residual TCN, BiLSTM and an attention mechanism to effectively capture temporal dependencies and discriminative EEG features. Initially, raw EEG signals undergo a preprocessing stage that comprises noise filtering, normalization, and segmentation to improve signal quality and ensure consistent input representation. The processed signals are then passed through the Residual TCN module, which utilizes dilated convolutions and residual connections to extract hierarchical temporal features from EEG sequences. The extracted attributes are subsequently analyzed by a BiLSTM layer to capture bidirectional temporal relationships inside the EEG data.

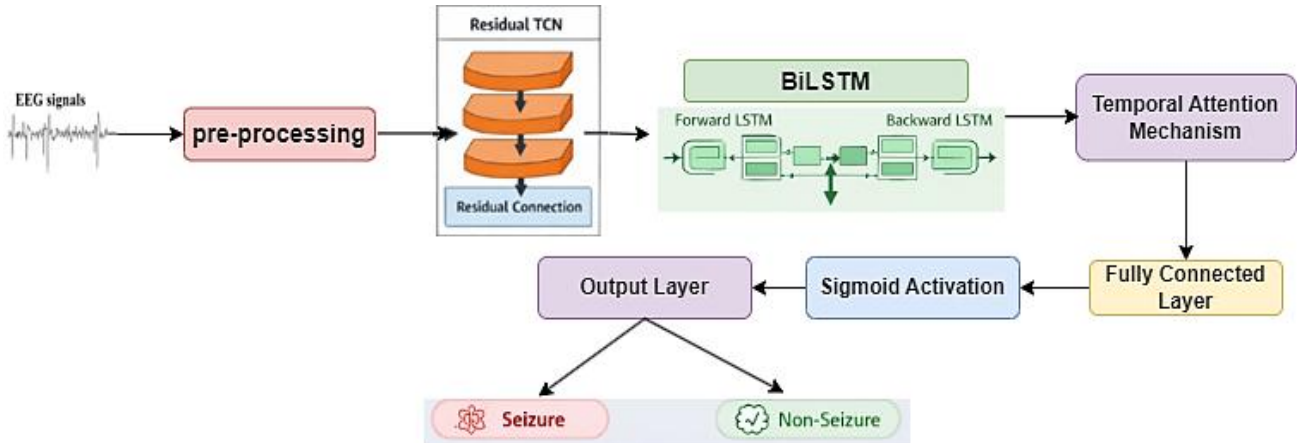


Figure. 1 Block diagram of proposed model

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Temporal attention mechanism is applied to disperse adaptive weights to informative variables and generate a context-aware representation of the EEG signals. Finally, a sigmoid activation succeeds after a fully connected layer is applied to the learned representations to perform binary classification of seizure and non-seizure EEG patterns. Fig. 1 depicts the architecture of the proposed Residual TCN-BiLSTM-Attention framework for EEG-based epileptic seizure detection.

3.1 Dataset description

Experimental analysis was performed using the publicly available Epileptic EEG Dataset from the Mendeley Data repository. The dataset contains EEG recordings collected from six individuals diagnosed with focal epilepsy between January 2014 and July 2015 at the American University of Beirut Medical Center's Epilepsy Monitoring Unit [23]. EEG signals were acquired using 21 scalp electrodes based on the international 10–20 electrode placement system and sampled at 500 Hz, enabling detailed capture of temporal brain activity associated with epileptic seizures.

During dataset preparation, channels Cz and Pz were excluded from the dataset due to inconsistencies and the presence of missing or noisy signal segments. The removal was applied uniformly across all samples to ensure consistent input dimensionality, without any channel interpolation or reconstruction, resulting in a fixed set of 19 EEG channels for all samples. All EEG recordings were verified to ensure that the remaining 19 channels were consistently available across all subjects. Samples with incomplete channel information were not included in the analysis. The channel ordering was fixed according to the standard 10–20 electrode configuration and maintained uniformly across all samples, ensuring consistent spatial representation. The dataset provides preprocessed and labeled EEG samples in the form of one-second segments, where each sample is structured as a matrix of size 19×500.

In total, the dataset contains 7,745 labeled EEG segments, including 3,895 non-seizure (interictal) and 3,850 seizure (ictal) samples. These standardized segments provide consistent temporal and spatial representations suitable for machine learning and deep learning-based epileptic seizure detection. Fig. 2 shows a sample snapshot of the dataset. The dataset is organized in a segment-level format, where multiple EEG segments are derived from the same subjects. All samples correspond to the pre-segmented one-second EEG segments provided by the dataset.

3.2 Data preprocessing

Preprocessing was performed to improve signal quality and prepare the EEG samples for effective DL-based epileptic seizure detection. Although the dataset provides pre-segmented EEG samples in a standardized format, additional preprocessing steps were applied to enhance data quality and model performance. EEG signals often contain noise, artifacts, and amplitude variations caused by environmental interference and physiological movements. Each EEG sample is represented as a temporal sequence of signal values, as expressed in Eq. (1).

Record Date	Gender	Age	Sampling rate	EEG Channels	File duration	File Name	Seizure type, focal point	File onset	File Offset	Nb of seizures	Seizure onset	Duration	Comments
9-Jul-15	F	5y	500	19	3:00:03	p10_Record1	Electrographic,Fp2/F8	5:36:39 AM	8:36:36 AM	1	7:36:38 AM	7 min, 25 sec	Channels Pz and Cz Removed due to artifact
10-Jul-15					3:00:06	p10_Record2	Electrographic,Fp2/F8	4:29:14 AM	7:29:07 AM	1	6:29:14 AM	5 min ,5 sec	Channels Pz and Cz Removed due to artifact
22-Oct-14	F	4.5 y	500	21	2:05:52	p11_Record1	CPS, Right temporal	1:58:50 PM	4:04:42 PM	1	3:08:55 PM	1 min 4 sec	
					2:50:28	p11_Record2	CPS, Right temporal	4:25:26 PM	7:15:46 PM	1	6:11:38 PM	51 sec	
					3:00:03	p11_Record3	CPS,T6 posterior temporal	6:53:21 PM	9:53:18 PM	4	7:18:38 PM	1 min 31 sec	
											8:06:38 PM	1 min 23 sec	
											8:53:22 PM	1 min 16 sec	
											9:27:24 PM	1 min 13 sec	
					3:00:03	p11_Record4	CPS,Fronto-temporal f4/f8	9:55:58 PM	12:55:56 AM	1	11:55:35 PM	22 min 38 sec	
3-Sep-14	M	8 y	500	21	3:00:03	p12_Record1	CPS/Diffuse onset	12:20:18 AM	3:20:16 AM	2	1:56:14 AM	1 min 16 sec	
											2:20:12 AM	1 min 44 sec	
					3:00:03	p12_Record2	CPS/Diffuse onset	3:50:53 AM	6:50:51 AM	2	5:50:55 AM	1 min 58 sec	
											6:40:30 AM	1 min 22 sec	Same seizure in two records, included once in labels
					2:34:22	p12_Record3	CPS/Diffuse onset	6:37:22 AM	9:11:39 AM	3	6:40:30 AM	1 min 22 sec	
											7:21:36 AM	2 min 44 sec	
											8:37:48 AM	1 min 53 sec	

Figure. 2 Sample dataset image

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$$X = \{x_1, x_2, \ldots, x_T\} \quad (1)$$

where x_t denotes the EEG signal value at time step t, and T represents the overall number of temporal samples within the signal segment.

To remove unwanted noise and retain relevant brain activity frequencies, a band-pass filtering operation with cutoff frequencies of 0.5 Hz to 40 Hz was applied to each EEG segment. The filtered signal $x_f(t)$ is defined as in Eq. (2).

$$x_f(t) = BPF(x(t)) \quad (2)$$

where $BPF(\cdot)$ denotes the band-pass filtering operation. The filtering was implemented using a fourth-order Butterworth filter, applied in a forward–backward manner to avoid phase distortion.

Following filtering, normalization was applied to stabilize the signal distribution and improve training convergence. Z-score normalization was applied in order to standardize the EEG signals. The normalized EEG signal x_t^{norm} is computed as in Eq. (3).

$$x_t^{norm} = \frac{x_t - \mu}{\sigma} \quad (3)$$

where μ signifies the mean value of the EEG signal and σ represents the standard deviation.

After normalization, the EEG samples were directly utilized as the fixed-length one-second segments provided by the dataset, without additional segmentation. Each segment corresponds to 500 temporal samples (1 second at 500 Hz) and is directly used as an input sample for the learning model, as represented in Eq. (4).

$$S_i = \{x_i, x_{i+1}, \ldots, x_{i+L-1}\} \quad (4)$$

where S_i signifies the ith EEG segment provided by the dataset and $L = 500$ represents the number of temporal samples in each segment. Finally, the EEG segments were organized into a structured input matrix suitable for DL-based temporal modeling. The input representation of the dataset is expressed in Eq. (5).

$$X_{input} \in R^{n \times L} \quad (5)$$

where n specifies the entire number of EEG segments and L signifies the number of time samples in each segment. This preprocessing pipeline ensured improved signal quality, normalized input representation, and structured temporal data.

In summary, the preprocessing pipeline consists of (i) band-pass filtering using a 0.5–40 Hz fourth-order Butterworth filter applied in a forward–backward manner, (ii) Z-score normalization applied independently to each EEG segment, and (iii) direct utilization of fixed-length one-second segments without additional segmentation. These steps ensure consistent and reproducible input representation across all samples.

3.3 Model development

A hybrid DL architecture was developed to detect epileptic seizures from EEG signals by integrating Residual TCN, BiLSTM, and a temporal attention mechanism. The Residual TCN was used to extract temporal structures from EEG recordings, while the BiLSTM layer captured bidirectional temporal dependencies within the signal patterns. An attention mechanism was incorporated to emphasize informative features for binary categorization of seizure and non-seizure EEG segments.

3.3.1. Residual TCN

The TCN is a DL framework specifically developed for modeling sequential and time-series data. Unlike conventional convolutional networks, which primarily process spatial features, TCN operates along the temporal dimension and is particularly effective for collecting temporal dependencies in sequential signals. One of the important characteristics of TCN is the usage of causal convolution, which confirms that the prediction at a specific time step in accordance to the present and preceding inputs without using future data [24]. This property preserves the temporal ordering of the signal and makes the architecture suitable for sequential data modeling. Fig. 3 displays the framework of the TCN.

To effectively capture long-range temporal dependencies, TCN retains dilated convolutions, which increase the receptive field of the structure by introducing gaps between filter elements. This mechanism enables the network to observe a wider temporal context without significantly increasing the number of parameters or computational complexity [25]. The dilated convolution applied to an input sequence $x(t)$ with filter $f(k)$ is represented in Eq. (6).

$$y(t) = \sum_{k=0}^{K-1} f(k)x(t - d \cdot k) \quad (6)$$

where $y(t)$ denotes the output sequence, K signifies the filter size, and d indicates dilation factor

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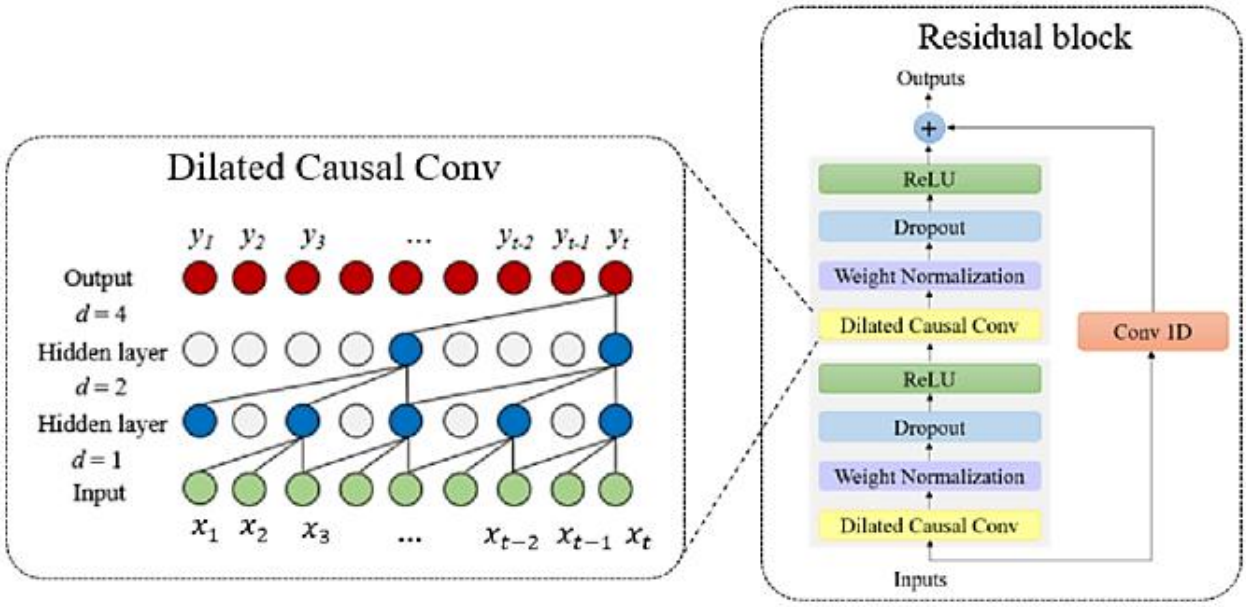


Figure. 3 Architecture of the TCN

controlling the spacing among convolutional operations. To enhance learning capability and prevent degradation in deep networks, residual connections are incorporated into the architecture. These connections allow the network to learn residual mappings by totalling the input of a block directly to its output. The residual mapping is mathematically expressed in Eq. (7).

$$H(x) = F(x) + x \tag{7}$$

where $F(x)$ signifies the transformation learned by the convolutional layers and x represents the data passed to the residual block.

In practice, a residual TCN block typically contains of dilated causal convolution layers succeeded by normalization and nonlinear activation functions, with a residual connection that facilitates stable gradient propagation and efficient learning of temporal patterns in sequential signals.

3.3.2. BiLSTM

RNNs are specialized neural networks for sequential data that maintain an internal state, allowing information from earlier inputs in the sequence to influence later predictions. Traditional RNNs can struggle to capture dependencies that span many time steps due to the disappearing gradient effect, which weakens the learning of long-range relations in sequences.

LSTM networks were developed to address this limitation of conventional RNN architectures. For a given input sequence x_t, the LSTM cell computes the forget gate, input gate, and output gate to regulate the update of the cell state and hidden state. The input gate regulates how much new information is written into the memory cell, as defined in Eq. (8).

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{8}$$

where i_t signifies the input gate at time step t, x_t signifies the input vector, h_{t-1} is the hidden state, W_i and U_i are weight matrices, and b_i is the bias term. The forget gate controls the amount of information from the previous cell that is preserved, as expressed in Eq. (9).

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \tag{9}$$

where f_t denotes the forget gate controlling the retention of previous memory, W_f, U_f, and b_f are learnable parameters. The candidate memory content is generated from the input and the previous hidden state, as computed in Eq. (10).

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \tag{10}$$

where $\tilde{c}_t$ represents the candidate cell state and $\tanh(\cdot)$ denotes the hyperbolic tangent activation function. The updated cell state is attained by combining the retained information and the newly generated candidate memory, as defined in Eq. (11).

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{11}$$

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where c_t denotes the updated cell state, c_{t-1} represents the previous cell state, and $\odot$ signifies element-wise multiplication. The output gate determines the hidden representation of the LSTM cell, which is calculated using Eq. (12) and Eq. (13).

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (12)$$

$$h_t = o_t \odot \tanh(c_t) \quad (13)$$

where o_t represents the output gate and h_t signifies the hidden state at time step t.

BiLSTM is an improved version of the conventional LSTM network developed to learn temporal dependencies in sequential data by analysing the input sequence in both forward and backward directions. By allowing the network to utilize contextual information from previous and upcoming time steps, this bidirectional processing improves the representation of temporal patterns in sequential signals. Fig. 4 illustrates the architecture of the BiLSTM network, depicting forward and backward information flow and hidden state propagation [26].

BiLSTM enhances the conventional LSTM architecture by evaluating the input sequence in forward and backward directions, which enables the network to learn contextual dependencies from past and future time steps. The forward and backward hidden states are calculated as given in Eq. (14) and Eq. (15).

$$\overrightarrow{h_t} = LSTM(x_t, \overrightarrow{h_{t-1}}) \quad (14)$$

$$\overleftarrow{h_t} = LSTM(x_t, \overleftarrow{h_{t+1}}) \quad (15)$$

where $\overrightarrow{h_t}$ and $\overleftarrow{h_t}$ denote the hidden states. The final contextual representation is acquired by concatenating the forward and backward hidden states, as defined in Eq. (16).

$$h_t = [\overrightarrow{h_t}; \overleftarrow{h_t}] \quad (16)$$

where $[\cdot;\cdot]$ denotes vector concatenation. This bidirectional learning mechanism enables the network to capture comprehensive temporal patterns in sequential signals, making BiLSTM particularly effective for modeling complex dependencies in EEG time-series data.

3.3.3. Temporal attention mechanism

Attention mechanisms are widely used in sequential learning models to emphasize the most informative features within temporal representations. The attention mechanism allocates the method to assign various importance weights to hidden representations generated by sequential networks, thereby enhancing the learning of discriminative temporal patterns. The sequence of hidden representations obtained from the previous sequential layer is denoted by $H = \{h_1, h_2, \ldots, h_T\}$. The attention score for each hidden state is computed as defined in Eq. (17).

$$e_t = v^T \tanh(W_h h_t + b_h) \quad (17)$$

where h_t signifies the hidden state at time step t, W_h denotes the weight matrix, b_h is the bias vector, and v is the learnable context vector.

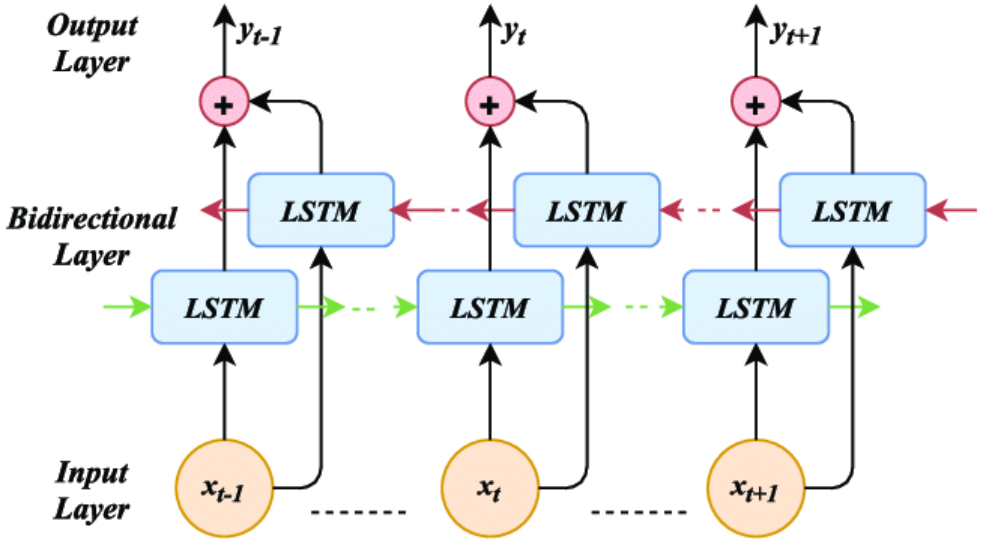


Figure. 4 Architecture of Bi-LSTM

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The attention weights are then normalized using the softmax function to attain a probability distribution over the hidden states, as expressed in Eq. (18).

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \quad (18)$$

where α_t represents the normalized attention weight assigned to the hidden state h_t, and T denotes the sum of the temporal stages in the sequence. Finally, the context vector representing the weighted temporal features is calculated as the weighted quantity of the hidden states, as defined in Eq. (19).

$$c = \sum_{t=1}^{T} \alpha_t \, h_t \quad (19)$$

where c denotes the final context vector obtained by aggregating the hidden states according to their attention weights. This system allows the framework to concentrate on essential temporal segments of EEG signals, thereby improving the representation of seizure-related patterns in sequential data.

3.3.4. Proposed TCN-BiLSTM-attention model

The proposed framework integrates Residual TCN, BiLSTM, and an attention mechanism to effectively model temporal dependencies in EEG signals. Initially, the preprocessed EEG signals are provided as input sequences to the Residual TCN layer, which performs temporal convolution using dilated convolutions to capture short- and long-range temporal dependencies. The TCN architecture consists of stacked residual blocks with a kernel size of 3 and dilation factors of 1, 2, and 4 to expand the receptive field while maintaining computational efficiency. Each residual block contains convolutional filters (64 filters), batch normalization, and ReLU activation to improve feature learning and training stability. The extracted temporal feature representations are then forwarded to the BiLSTM layer with 128 hidden units to capture bidirectional temporal dependencies present in EEG sequences. The forward and backward LSTM layers process the sequential EEG features simultaneously, generating contextual representations that incorporate both past and future temporal information. These hidden representations are subsequently passed to an attention layer, which computes importance weights for each temporal step and generates a context vector representing the most informative EEG segments associated with seizure activity. The context vector produced by the temporal attention mechanism is fed into a dense layer with 64 neurons, succeed by a sigmoid activation function to distinguish between seizure and non-seizure conditions. Training is conducted over 100 epochs with 32 of batch size, which promotes efficient convergence and reduces computational cost. This integrated architecture allows the Residual TCN to capture hierarchical temporal patterns, the BiLSTM to model bidirectional dependencies, and the attention mechanism to emphasize informative EEG segments, thereby improving the overall capability of the framework in identifying epileptic seizure events from EEG signals. The output layer produces a probability value demonstrating whether the given EEG segment corresponds to a seizure or non-seizure event. A sigmoid activation is applied to generate the output probability $\hat{y}$, where values 0 and 1 indicate non-seizure and seizure EEG patterns, respectively. The algorithm for the proposed framework is given below.

Algorithm: Residual TCN-BiLSTM-Attention Framework for EEG Seizure Detection

Input: Preprocessed EEG signal matrix $X \in \mathbb{R}^{T \times C}$
Output: Predicted seizure label (Seizure / Non-Seizure)

Begin

Step 1: Data Preprocessing

- Load EEG signal dataset
- Remove noise and artifacts using filtering techniques
- Normalize EEG signal values for stable training
- Segment EEG recordings into fixed-length temporal windows
- Represent each EEG segment as an input feature sequence X

Step 2: Residual TCN Feature Extraction

- Initialize Residual TCN parameters:
  - Kernel size $K = 3$
  - Filters = 64
  - Dilation factors $d = \{1,2,4\}$
- For each residual block $i = 1 \ldots N$
  - Apply dilated convolution
  - Apply Batch Normalization
    $$\tilde{y}(t) = BN(y(t))$$
  - Apply ReLU activation
    $$h(t) = ReLU(\tilde{y}(t))$$
  - Apply residual connection
    $$H(x) = F(x) + x$$
- Obtain temporal feature representation
  $$F_{TCN}$$

Step 3: Sequence Representation

- Convert extracted TCN features into ordered temporal sequence

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$$F_{TCN} \in \mathbb{R}^{T \times d}$$

Step 4: Bidirectional Temporal Dependency Modeling

- Input the extracted TCN features into the BiLSTM layer.
- Compute forward and backward hidden states

$$h_t^{forward} = LSTM(X_t, h_{t-1})$$
$$h_t^{backward} = LSTM(X_t, h_{t+1})$$

- Concatenate hidden states to obtain contextual representation

$$H_t = [h_t^{forward} \parallel h_t^{backward}]$$

Step 5: Temporal Attention Mechanism

- Evaluate attention score for each hidden state

$$e_t = v^T \tanh(W_h H_t + b_h)$$

- Apply the softmax function to normalize the attention weights.

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}$$

- Generate weighted context vector

Step 6: Classification Layer

- Feed the context vector into a fully connected layer

$$z = W_c c + b_c$$

- Compute probability score using sigmoid activation

$$\hat{y} = \sigma(z)$$

Step 7: Model Inference

- Feed the preprocessed EEG signal segments into the trained Residual TCN-BiLSTM-Attention model.
- The method processes the input features and generates a probability score using the sigmoid output layer.
- Apply a decision threshold of 0.5 to convert the probability into a binary label.

Step 8: Output

- Output the final predicted label (Seizure / Non-Seizure).

End

3.4 Simulation setup

The proposed Residual TCN-BiLSTM-Attention framework was executed and evaluated in a computing environment designed to support DL experiments efficiently. All simulations were completed on a workstation installed by an Intel Core i7 processor, NVIDIA GeForce GPU, and 16 GB RAM to handle the computational requirements of the hybrid neural network framework. The model was developed using Python with the TensorFlow framework and the Keras API, which enables flexible construction of DL models that combine convolutional, recurrent, and attention-based components. The EEG dataset was preprocessed and organized into sequential segments before being provided as input to the model. Residual TCN blocks were employed to capture temporal feature representations, while the BiLSTM layer modelled sequential dependencies in the EEG signals. The model training process involved tuning key hyperparameters to achieve stable convergence and reliable classification performance. The final hyperparameter settings adopted in this framework are presented in Table 1.

In addition to the hyperparameters listed, further implementation details are provided to ensure reproducibility. The input EEG data are arranged as tensors of shape (samples × time steps × channels), where each sample corresponds to a segment of 500-time steps with 19 channels. The Adam optimizer was configured with a learning rate of 0.0001 and default momentum parameters ($\beta1$=0.9, $\beta2$=0.999). Model weights were initialized using Glorot uniform initialization. Early stopping based on validation loss was applied to prevent overfitting, and the best model weights were retained using a checkpoint mechanism. A fixed random seed (42) was used to ensure consistency across experiments. The dataset split (80:20 train–test) was performed using the same seed to ensure reproducibility across all experiments, and all baseline models and the proposed model were trained and evaluated using the same preprocessing pipeline and data partitioning. The reported results correspond to the best-performing model obtained during training. For model evaluation, the segment-level data split implies that EEG segments derived from the same subject may be distributed across both training and testing sets. As a result, the evaluation protocol does not represent a subject-independent setting and may introduce data leakage due to shared subject-specific characteristics across splits.

Table 1. Hyperparameters of Residual TCN-BiLSTM-Attention model

Parameter	Value
Number of Epochs	100
Optimizer	Adam
Dropout Rate	0.3
Loss Function	Binary Cross-Entropy
Batch Size	32
Learning Rate	0.0001
Activation Function	Sigmoid

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The combination of fixed preprocessing steps, clearly defined hyperparameters, and controlled data partitioning ensures that the proposed framework can be replicated without ambiguity.

4. Results and discussions

4.1 Performance evaluation

Accuracy and loss plots illustrate the learning performance of a deep learning model during training by showing how prediction accuracy increases and error (loss) decreases over successive epochs. Fig. 5 demonstrates the training and validation accuracy curves of the Residual TCN-BiLSTM-Attention model during the training process. The training accuracy gradually rises through the number of epochs, indicating that method effectively learns discriminative temporal patterns from the EEG signals. At the initial stage, the accuracy begins around 89-90% and steadily improves as training progresses. The validation accuracy follows a similar trend, demonstrating stable learning behavior and effective generalization capability. Both curves show consistent improvement during the early training phase and stabilize around 40-50 epochs, while training continues until the maximum epoch limit of 100 epochs. At the final stage, the model achieves a training accuracy of approximately 98.9%.

The training and validation loss trends of the suggested Residual TCN-BiLSTM-Attention model over 100 training epochs are presented in Fig. 6.

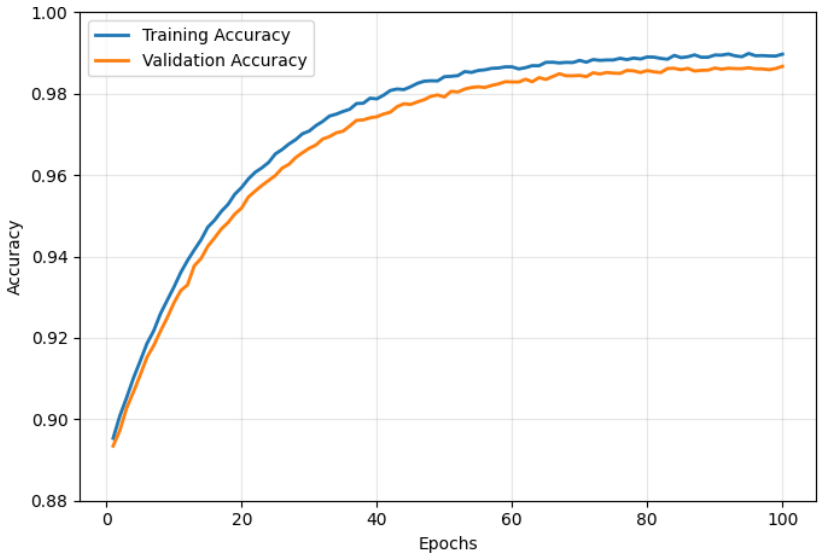


Figure. 5 Accuracy plot of the proposed model

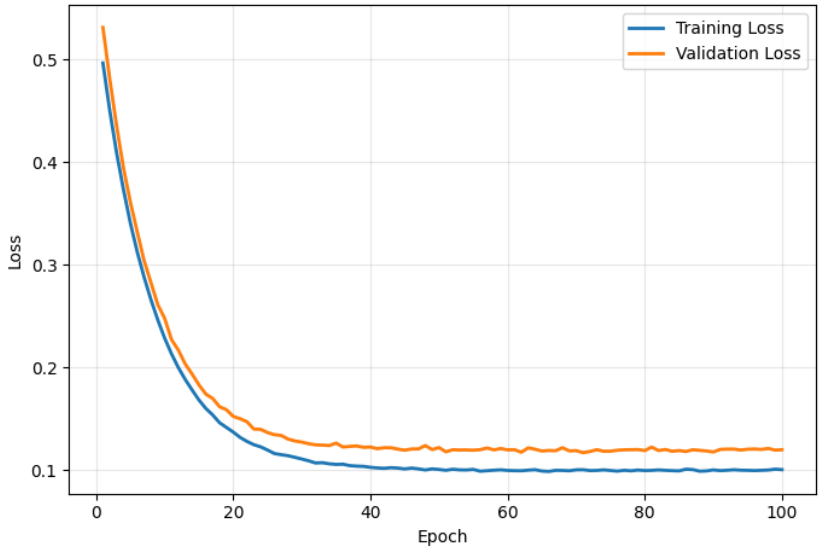


Figure. 6 Loss plot of the proposed model

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The training loss declines speedily through the initial epochs, indicating that the method effectively acquires discriminative temporal features from the EEG signals. A similar decreasing trend is observed in the validation loss, demonstrating stable learning behavior and consistent generalization capability. Around the 30-40 epochs range, both curves gradually stabilize. Throughout the training process, the validation loss is slightly greater than the training loss, which is typical in DL models and indicates that the model does not exhibit significant overfitting. The training loss approaches approximately 0.10, while the validation loss stabilizes around 0.12, confirming the robustness and learning stability of the proposed framework.

The effectiveness of the method was estimated by means of classification measures, including precision, accuracy, recall, and F1-score. This evaluation measures deliver a comprehensive valuation of the framework's predictive performance by analyzing overall classification correctness, reliability of seizure predictions, sensitivity in identifying seizure events, and the balanced trade-off among precision and recall. Together, these metrics delivers a reliable valuation of the method's ability to accurately differentiate among seizure and non-seizure EEG signals. The mathematical formulations of these evaluation metrics are expressed in Eqs. (20)-(23).

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \tag{20}$$

$$\text{Recall} = \frac{TP}{TP+FN} \tag{21}$$

$$\text{Precision} = \frac{TP}{TP+FP} \tag{22}$$

$$\text{F1-score} = \frac{2TP}{2TP+FP+FN} \tag{23}$$

In the above expressions, TP, TN, FP, and FN signify true positives, true negatives, false positives, and false negatives, respectively. The framework attains consistently high precision, recall, and F1-score values, indicating effective identification of epileptic seizure patterns in EEG signals. The overall classification accuracy of 98.8% further confirms the robustness, reliability, and efficacy of the model for accurate epileptic seizure identification. These results represent the primary performance of the proposed model obtained using an 80:20 hold-out train–test split. Table 2 denotes the performance assessment of the developed framework.

Fig. 7 illustrates the confusion matrix of the seizure detection framework for the binary category of EEG signals into seizure and non-seizure classes. Using an 80:20 train-test split, 80% of the dataset (6,196 samples) was allocated for training and 20% (1,549 samples) remaining was reserved for testing.

Table 2. Performance evaluation of the Residual TCN-BiLSTM-Attention model

Metric	Value
Accuracy	98.8%
Recall	98.8%
Precision	98.7%
F1-score	98.8%
AUC	0. 998

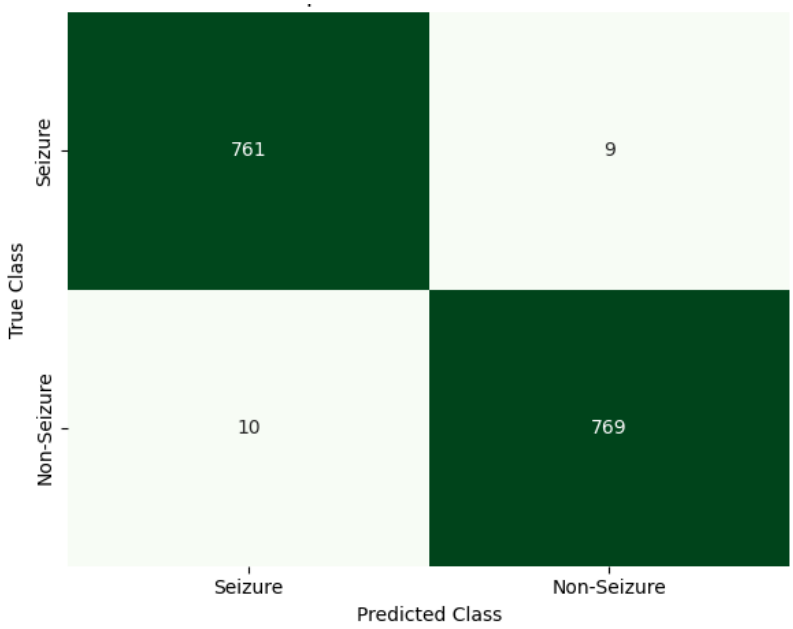


Figure. 7 Confusion matrix of the Residual TCN-BiLSTM-Attention model

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From the testing set, the method correctly identified 761 seizure instances as seizure (true positives) and 769 non-seizure instances as non-seizure (true negatives). However, 9 seizure samples were incorrectly classified as non-seizure (false negatives), and 10 non-seizure samples were misclassified as seizure (false positives). The confusion matrix demonstrates that the proposed framework effectively distinguishes among seizure and non-seizure EEG signals with a high level of accuracy, indicating better classification performance for epileptic seizure detection.

The ROC curve of the framework for epileptic seizure is illustrated in Fig.8. The ROC curve of the model lies adjacent to the upper-left corner of the plot, representing a higher capability to correctly distinguish seizure and non-seizure EEG segments. The dashed diagonal line characterizes the performance of a random classifier, while the proposed model significantly outperforms this baseline. The computed AUC value of 0.998 demonstrates excellent discriminative performance and confirms the efficiency of the suggested method in identifying epileptic seizure patterns. The high AUC value indicates that the model maintains a high detection rate while keeping false positive predictions minimal, thereby ensuring reliable seizure classification.

4.2 Repeated-run performance analysis

To evaluate statistical stability, the proposed model was executed over multiple independent runs under identical experimental conditions. The results, presented in Table 3, show consistent performance with accuracy values ranging from 98.6% to 98.9%. The mean accuracy of 98.76% with a standard deviation of ±0.114 indicates low variability across runs.

This stability demonstrates that the proposed Residual TCN-BiLSTM-Attention architecture produces reliable predictions and is not significantly affected by random initialization or stochastic training dynamics. The consistent behavior across runs further suggests that the integration of residual temporal convolution, bidirectional sequence modeling, and attention mechanisms contributes to robust feature learning and stable convergence during training. Overall, the low variance in performance confirms the reliability of the proposed framework. All reported performance metrics in this study are derived from the average of these repeated runs to ensure reproducible evaluation.

Table 3. Repeated-run performance of the proposed model

Run	Accuracy (%)
Run 1	98.7
Run 2	98.9
Run 3	98.8
Run 4	98.6
Run 5	98.8
Mean ± Std	98.76 ± 0.114

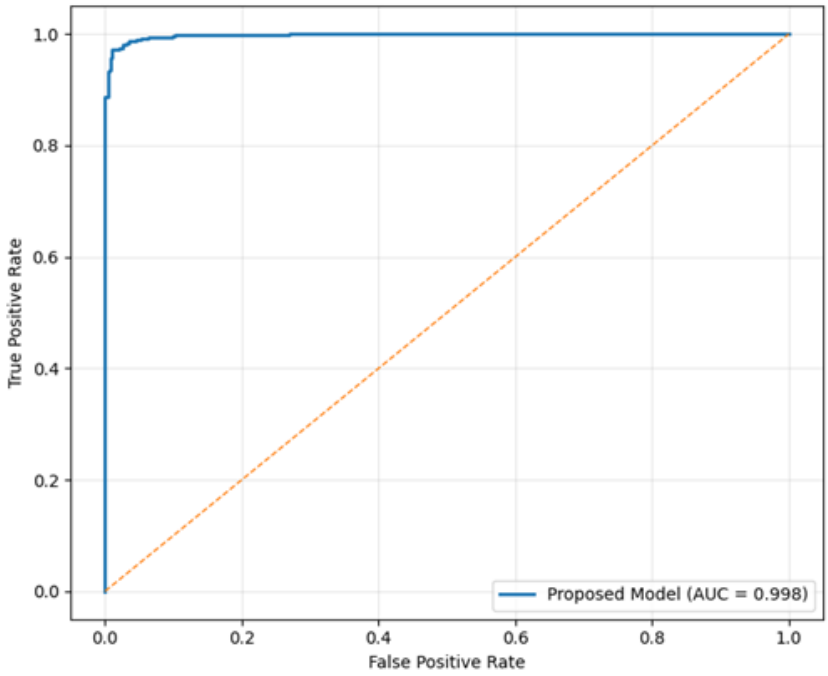


Figure. 8 ROC curve of proposed Residual TCN-BiLSTM-Attention model

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4.3 Statistical significance analysis

To further assess statistical significance, a one-sample one-tailed t-test was conducted on the repeated-run accuracy values to evaluate whether the observed mean classification accuracy exceeds a reference value of μ_0 =98.0%, which serves as an indicative benchmark close to the performance of baseline models evaluated in this study. The null hypothesis assumes that the mean classification accuracy is equal to μ_0 =98.0%. The t-statistic is computed as:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x}$ denotes the mean accuracy, μ_0 represents the reference (null) mean, s is the sample standard deviation, and n is the number of repeated runs.

The repeated-run accuracies are 98.7%, 98.9%, 98.8%, 98.6%, and 98.8%, yielding a mean accuracy of 98.76% with a standard deviation of 0.114%, indicating low variability and stable convergence of the model. Substituting these values:

$$t = \frac{98.76 - 98.0}{0.114/\sqrt{5}} \approx 14.9$$

With n = 5 runs, the degrees of freedom are n − 1 = 4. The corresponding p-value obtained from the t-distribution is less than 0.0001, suggesting that the observed mean accuracy exceeds the reference value under the defined experimental conditions. Table 4 summarizes the statistical test results.

However, since repeated runs were performed on the same dataset partition, strict independence among samples is not guaranteed. Therefore, the one-sample t-test is considered indicative rather than conclusive. To provide a more reliable evaluation, cross-validation results are presented in the subsequent section, where the model demonstrates consistent performance across folds, indicating robust and stable behavior across different data splits.

4.4 Ablation study

Ablation study was conducted to evaluate the involvement of every element in the Residual TCN-BiLSTM-Attention architecture for epileptic seizure detection. In this analysis, several model variants were examined by selectively modifying or removing individual modules while keeping the remaining network structure unchanged. Initially, a baseline configuration consisting only of the Residual TCN was evaluated to assess its ability to identify temporal characteristics from EEG signals. Subsequently, the BiLSTM layer was incorporated to capture bidirectional temporal dependencies within the EEG sequences. Finally, the attention mechanism was integrated to disperse adaptive importance weights to the learned representations and emphasize the most informative temporal segments. The results reported in this section are obtained using the same 80:20 hold-out train–test split applied for the primary evaluation.

The comparative outcomes of these configurations are given in Table 5. The outcomes indicate that each architectural component contributes to improved detection performance, with the complete Residual TCN-BiLSTM-Attention model achieving the highest classification accuracy. The addition of the BiLSTM layer enhances sequential pattern learning, while the attention mechanism improves the model's ability to highlight discriminative EEG features associated with seizure activity. Overall, the ablation analysis confirms that the combined architecture provides a more effective representation of EEG signals, leading to improved seizure classification performance. The low variation observed across repeated runs further supports the reliability of the performance improvements contributed by each architectural component.

4.5 Cross-validation protocol

To further examine the reliability and stability of the proposed Residual TCN-BiLSTM-Attention architecture, a 5-fold cross-validation approach was applied.

Table 4. Statistical significance analysis (one-sample t-test)

Metric	Value
Mean accuracy (%)	98.76 ± 0.114
Null mean (μ_0)	98.0
t-statistic	14.9
p-value	< 0.0001

Table 5. Ablation study results

Model Variant	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	ROC-AUC
Residual TCN Only	96.9	96.5	96.7	96.6	0.97
Residual TCN + BiLSTM	97.8	97.4	97.6	97.5	0.98
Residual TCN + BiLSTM + Attention (Proposed)	98.8	98.7	98.8	98.8	0.998

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This evaluation is conducted independently of the hold-out testing and is intended to assess model stability across multiple data splits. The entire EEG dataset was divided into five equal partitions for cross-validation. For model evaluation, the dataset was randomly partitioned at the segment level into training, validation, and testing subsets, where each fold was used once as the validation set while the remaining folds were used for training. The data partitioning was performed using a fixed random seed (42) to ensure reproducibility of the experimental results.

The outcomes attained from the cross-validation study are summarized in Table 6. The proposed framework maintained consistent performance across the different folds, with classification accuracy values ranging between 98.2% and 99.0%. The mean accuracy obtained from the five iterations was 98.62% (approximately 98.6% after rounding), accompanied by high precision, recall, and F1-score values. The limited variation in performance across folds indicates stable learning behavior and robustness of the model under repeated training conditions.

However, it is important to explicitly emphasize that cross-validation was performed at the segment level rather than at the subject level. As multiple EEG segments are derived from a limited number of subjects, segments from the same subject are likely to be present in both training and testing subsets. This results in an inherent risk of data leakage, where the model may capture subject-specific patterns instead of fully generalized seizure characteristics. Consequently, the adopted evaluation protocol does not represent a subject-independent setting.

Therefore, the reported cross-validation performance should be interpreted as segment-level classification capability under intra-subject conditions. While the results demonstrate the effectiveness of the proposed architecture in capturing discriminative temporal patterns within the dataset, they may overestimate performance in scenarios involving unseen patients. Despite this limitation, the inclusion of cross-validation, repeated-run analysis, and statistical significance testing provides evidence of the model's stability and consistent behavior across different data partitions under the defined experimental setup.

Table 6. Performance results of 5-fold cross-validation for the proposed Residual TCN-BiLSTM-Attention model

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	ROC-AUC
Fold 1	98.2	98.0	98.1	98.0	0.986
Fold 2	98.5	98.3	98.4	98.3	0.988
Fold 3	98.8	98.6	98.7	98.6	0.991
Fold 4	99.0	98.8	98.9	98.8	0.989
Fold 5	98.6	98.4	98.5	98.4	0.988
Average	98.6	98.4	98.5	98.4	0.988

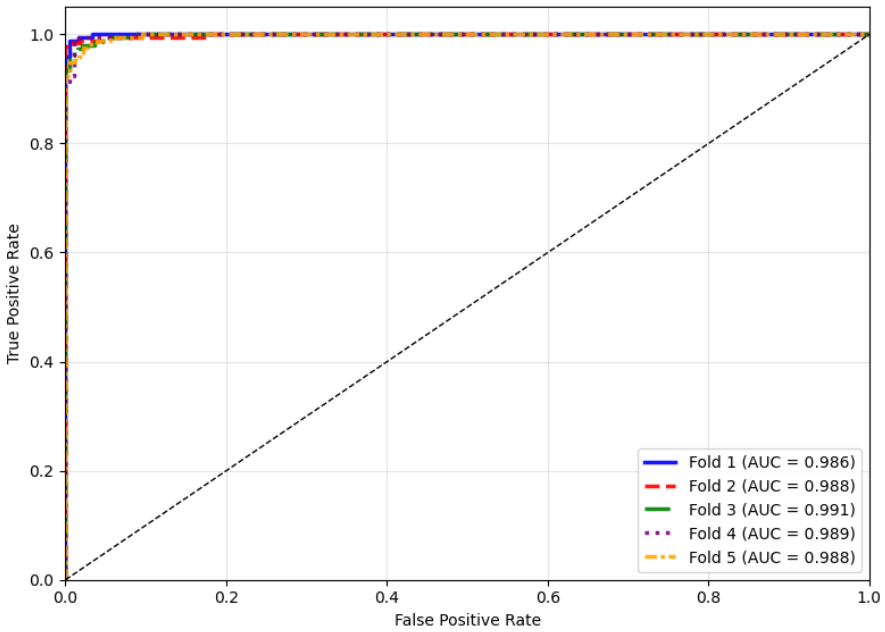


Figure. 9 ROC of 5 folds

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All reported performance metrics correspond to the average results obtained from multiple independent training runs conducted using different random initializations, thereby reducing variability associated with stochastic training and ensuring reliable performance estimation. This evaluation protocol is consistent with several prior EEG-based seizure detection studies that adopt segment-level validation, although it remains less strict than subject-independent evaluation. The reported cross-validation performance metrics are consistent with the repeated-run analysis, where results are obtained as the average of multiple independent runs to ensure reliable and reproducible performance estimation.

4.6 Comparison with baseline models

The efficacy of the Residual TCN-BiLSTM-Attention framework was further examined through comparison with several commonly used baseline models for EEG-based epileptic seizure detection. In this evaluation, representative machine learning and deep learning approaches, including SVM, RF, CNN, and LSTM, were implemented using the same dataset and consistent experimental settings to ensure a fair comparison.

To ensure reproducibility, the baseline models were implemented with explicitly defined configurations. The SVM model utilized a radial basis function (RBF) kernel with regularization parameter C=1 and kernel coefficient γ=0.01. The RF model consisted of 100 decision trees with a maximum depth of 10 to balance bias and variance.

The CNN model included two convolutional layers (32 and 64 filters, kernel size 3), each followed by max-pooling and a fully connected layer for classification. The LSTM model consisted of a single LSTM layer with 128 hidden units followed by a dense output layer. For deep learning models, the Adam optimizer, batch size of 32, binary cross-entropy loss function, and 100 training epochs were used. Hyperparameters were selected based on commonly adopted configurations in the literature and further refined through empirical tuning to ensure stable convergence. To ensure fairness in comparison, hyperparameters for all baseline models were selected using the same training–validation split and subjected to a comparable level of empirical tuning. The same preprocessing pipeline, data partitioning, and training conditions were applied consistently across all models. While the proposed architecture incorporates additional components such as residual temporal convolution and attention mechanisms, the baseline models were configured with comparable model capacities and trained under similar optimization settings. Differences in architectural complexity are inherent to the model design and are acknowledged as part of the comparative evaluation.

The comparative results summarized in Table 7 indicate that the proposed model achieves higher classification accuracy compared to baseline models under the same experimental conditions. Conventional machine learning models such as SVM and RF provide moderate detection accuracy due to their limited ability to capture complex temporal dependencies in EEG signals. Deep learning models such as CNN and LSTM demonstrate improved performance by learning hierarchical and sequential representations. The integration of Residual Temporal Convolutional Networks, BiLSTM, and attention mechanisms enables the proposed framework to learn more discriminative EEG representations, contributing to improved classification performance within the defined experimental setting. Fig. 10 illustrates the performance comparison between baseline models and the proposed framework.

4.7 Comparison with existing studies

Table 8 and Fig. 11 present the accuracy comparison between the proposed Residual TCN-BiLSTM-Attention model and several existing epileptic seizure detection approaches. Previous studies have employed a range of machine learning and deep learning architectures, including XGBoost-LSTM, RNN, CNN-BiGRU, and CNN-GRU models for EEG signal classification. The XGBoost-LSTM framework achieved an accuracy of 96.80%, with performance influenced by its reliance on handcrafted feature extraction and hybrid modeling.

Table 7. Performance comparison between the proposed model and baseline models for epileptic seizure detection using EEG signals

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
SVM	93.8	93.1	93.4	93.2
RF	95.6	95.1	95.3	95.2
CNN	96.9	96.4	96.6	96.5
LSTM	97.5	97.1	97.3	97.2
Residual TCN-BiLSTM-Attention (Proposed)	**98.8**	**98.7**	**98.8**	**98.8**

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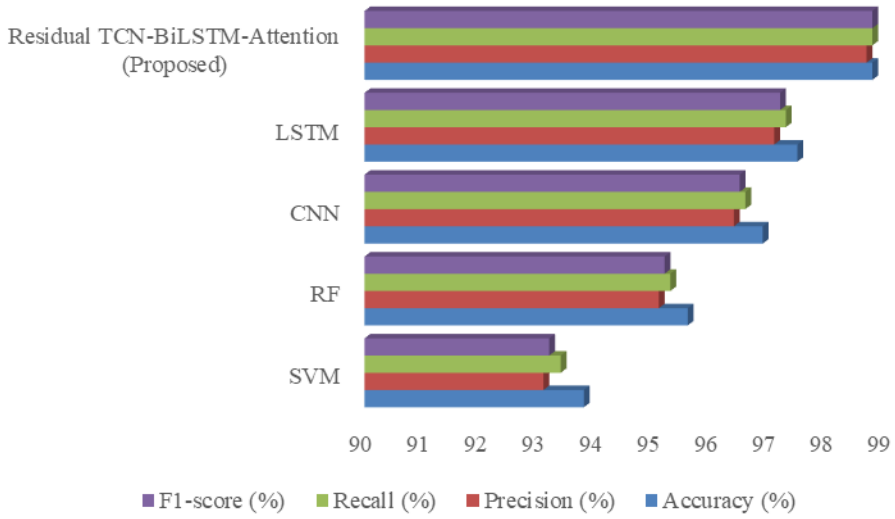


Figure. 10 Visualization of baseline comparisons

Table 8. Accuracy comparison of the suggested model with existing approaches

Study	Method	Dataset	Accuracy (%)
Betgeri et al. [14]	XGBoost + LSTM	EEG dataset	96.80
kina et al. [19]	RNN	EEG dataset	97.00
Paneru [8]	CNN-BiGRU	Mendeley EEG dataset	97.43
Bhadra et al. [18]	CNN-GRU	UCI + Mendeley dataset	97.50
Proposed Model	**Residual TCN-BiLSTM-Attention**	**Mendeley EEG dataset**	**98.80**

Note: The comparison in Table 8 is based on performance values reported in the literature review (Section 2), where different studies utilize varying EEG datasets and experimental settings. Due to the limited availability of standardized public datasets, direct comparison under identical conditions is not feasible; therefore, this comparison serves as a qualitative reference rather than a strict quantitative evaluation.

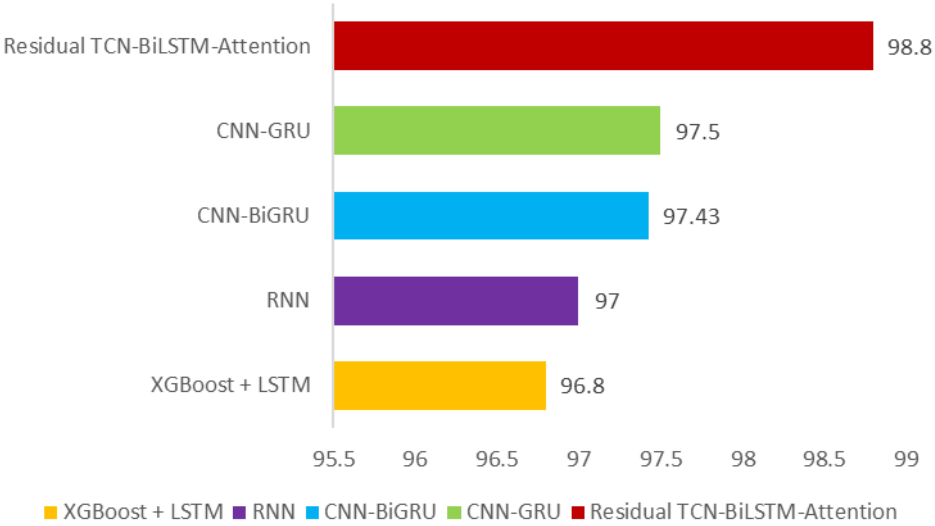


Figure. 11 Graphical illustration of accuracy comparison of the proposed model with existing methods

The RNN-based approach reported an accuracy of 97.00%, although it utilized a single sequential structure without advanced temporal feature extraction capabilities. Hybrid models such as CNN-BiGRU and CNN-GRU achieved accuracies of 97.43% and 97.50%, respectively; however, these approaches do not incorporate mechanisms for multi-level temporal interaction or attention-based feature refinement.

In comparison, the proposed Residual TCN-BiLSTM-Attention model achieved an accuracy of 98.80% on the Mendeley EEG dataset under the defined experimental setup. This result indicates the effectiveness of the proposed architecture in

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capturing temporal dependencies and discriminative EEG features within the dataset. The integration of Residual TCN, BiLSTM, and attention mechanisms improves feature representation and contributes to enhanced classification capability. However, this performance should be interpreted within the context of the adopted validation protocol and dataset characteristics rather than as a direct indicator of superiority over existing approaches.

It is important to emphasize that the comparative accuracy values reported in Table 8 are influenced by differences in validation strategies, dataset composition, and experimental configurations across studies. In particular, several existing works employ subject-wise or more rigorous validation protocols, whereas the present study adopts segment-level evaluation. Therefore, direct comparison of accuracy values should be interpreted cautiously, and the higher accuracy obtained in this study does not necessarily imply improved generalization across unseen subjects.

Although some of the compared studies utilize the same dataset, variations in experimental configurations and evaluation protocols limit direct comparability. The comparative analysis therefore includes both same-dataset and cross-dataset studies, and the reported performance differences should be interpreted with consideration of dataset-specific and methodological variations. Overall, the proposed framework demonstrates competitive performance within the defined experimental setting.

The evaluation presented in this study is conducted on a publicly available EEG dataset comprising a limited number of subjects and focuses on binary classification of seizure and non-seizure segments. While the proposed framework demonstrates strong performance under the given experimental setup, the findings should be interpreted within the scope of this dataset. The absence of subject-independent validation, such as leave-one-subject-out evaluation, and the lack of external dataset validation limit the ability to assess generalization across diverse patient populations, recording conditions, and clinical environments. Therefore, the proposed framework should be considered as a proof-of-concept for EEG-based seizure classification rather than a fully validated clinical deployment solution.

5. Conclusion

This study presented a hybrid deep learning framework for epileptic seizure detection from EEG signals using a Residual TCN-BiLSTM-Attention architecture. The proposed framework integrates residual temporal convolution to capture local temporal dependencies, bidirectional recurrent learning to model long-range sequential patterns, and a temporal attention mechanism to emphasize the most relevant EEG features. The effectiveness of the framework was evaluated using the Mendeley EEG dataset, which contains multichannel recordings of seizure and non-seizure brain activity. Experimental results indicated that the model achieves an accuracy of 98.8%, along with high recall, precision, and F1-score values under the defined experimental setup. ROC analysis further demonstrated strong discriminative capability between seizure and non-seizure EEG segments within the dataset. Comparative evaluation with existing methods shows that the proposed framework provides competitive performance under similar experimental conditions. However, the evaluation is conducted using segment-level data partitioning on a dataset with a limited number of subjects. The absence of subject-wise validation (e.g., leave-one-subject-out), patient-level cross-validation, and external dataset evaluation limits the assessment of subject-independent generalization. Due to this segment-level evaluation protocol, the results may overestimate the model's generalization performance when applied to completely unseen subjects. Therefore, the reported performance should be interpreted as indicative of model capability under controlled experimental conditions and not as definitive evidence of subject-independent robustness or real-world clinical applicability without further validation. Overall, the study presents an empirical evaluation of a hybrid deep learning architecture for EEG-based seizure classification, demonstrating its effectiveness under the defined experimental conditions and providing a foundation for further investigation under more rigorous validation settings.

Conflicts of Interest

The authors declare no conflict of interest.

Author Contributions

The paper conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, visualization, have been done by 1st author. The supervision and project administration, have been done by 2nd author.

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Acknowledgments

We would like to express our sincere gratitude to all those who contributed to the completion of this research paper. We extend our heartfelt thanks to family, colleagues and fellow researchers for their encouragement and understanding during the demanding phases of this work.

List of Notations

Symbol	Description
N	Number of samples in a segment
x_t	EEG signal at time step t
$\hat{x}$	Filtered EEG signal
μ	Mean of the signal
σ	Standard deviation
S_i	EEG segment
W	Temporal window length
h_t	Hidden state
c_t	Cell state
α_t	Attention weight
y	Output prediction

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